

Matthew Osgood

Data Engineer

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SUMMARY

Data Engineer with 3+ years of experience building production ETL pipelines, REST APIs, and large-scale data processing systems in financial services. Specialized in PySpark, Microsoft Fabric, and Delta Lake architectures across bronze/silver/gold lakehouse layers. Proven track record of delivering automation that drives measurable cost savings (~\$150K/month) and shipping full-stack products end-to-end. Comfortable across the full data and software stack — from SQL Server stored procedures to Next.js applications.

EXPERIENCE

Database & Reporting Developer — Planet Home Lending

July 2025 – Present

- Build and maintain PySpark pipelines on Microsoft Fabric processing mortgage and loan-servicing data across bronze, silver, and gold lakehouse layers.
- Designed and refactored SCD Type 2 notebooks for Encompass loan data, including custom column sanitization to handle numeric and special-character source fields.
- Optimized a high-cost Bronze refresh pipeline by eliminating 59 Spark cold starts, materially reducing CU consumption on an F64 capacity.
- Built Gold warehouse ETL notebooks using COPY INTO and MERGE patterns to incrementally load curated tables for downstream reporting.
- Authored SQL Server stored procedures (e.g., procLFNLoanIndicator) for dual Mercury and SFTP file output, plus a PHL Retail transaction feed pipeline.
- Hosted open-source LLMs (Llama 3) and built RAG pipelines with embeddings and vector search to power internal automation and decision workflows.

Back End Developer — Planet Home Lending

July 2023 – July 2025

- Engineered Python and SQL ETL pipelines plus REST APIs to automate loan-servicing workflows at scale.
- Designed scalable backend systems to process large loan transfers reliably with strong data integrity guarantees.
- Automated compliance checks, letter generation, and onboarding tasks using JAMS Scheduler and integrated APIs.
- Drove automation initiatives that reduced operational costs by approximately \$150K per month.

SKILLS

- **Languages:** Python, SQL, TypeScript, JavaScript, T-SQL
- **Data Engineering:** PySpark, Delta Lake, Microsoft Fabric, ETL/ELT, Lakehouse (Bronze/Silver/Gold), SCD Type 2, MERGE / COPY INTO, SQL Server, PostgreSQL
- **Backend & APIs:** REST APIs, Django REST, Next.js API routes, Stripe integration, JAMS Scheduler
- **Cloud & DevOps:** AWS (SES), Supabase, Vercel, GitHub, CI/CD, Linux / WSL, Docker
- **AI / LLMs:** RAG, embeddings, vector search, prompt engineering, fine-tuning, Llama 3
- **Other:** Systems design, performance optimization, workflow automation

PROJECTS

Sell Local — Founder & Full-Stack Developer selllocal.app

- Built and operate a SaaS platform for cottage-food businesses with storefronts, Stripe payments, pickup ordering, a notifications hub, bio-link pages, and digital product sales.
- Architected the full stack end-to-end: TypeScript / Next.js frontend, Supabase backend, Stripe Connect payments, and AWS SES transactional email.
- Run growth and lifecycle marketing including Meta ad campaigns and SEO, with live customer deployments in production.

Molloy Foundation Tournament Platform — Full-Stack Developer

- Developed a Next.js + Supabase + Stripe tournament registration platform with real-time updates and secure authentication for an annual charity basketball event.

EDUCATION

B.A. Economics & Data Analytics — University of Connecticut

August 2018 – May 2022

Coursework: Machine Learning, Data Visualization, Statistical Modeling (Python), Business Information Systems (RDBMS).

CERTIFICATES & CONTINUING EDUCATION

- DeepLearning.AI — RAG, Prompt Engineering, Fine-Tuning LLMs, Hugging Face, Open-Source LLMs, PyTorch.
- Coursera — Python Automation, API Development.